4 Physical Layer
4.3 Copper Cabling
4.3.1 Characteristics of Copper Cabling
4.3.2 Types of Copper Cabling
4.3.3 Unshielded twisted-pair (UTP)
4.3.4 Shielded twisted-pair (STP)
4.3.5 Coaxial cable
4.3.6 Check Your Understanding - Copper Cabling
4.4 UTP Cabling
4.5 Fiber-Optic Cabling
4.6 Wireless Media
4.7 Module Practice and Quiz
5 Number Systems
6 Data Link Layer
7 Ethernet Switching

Copper Cabling

4.3.1

Characteristics of Copper Cabling

Copper cabling is the most common type of cabling used in networks today. In fact, copper cabling is not just one type of cable. There are three different types of copper cabling that are each used in specific situations.

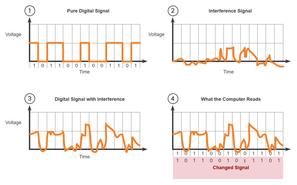
Networks use copper media because it is inexpensive, easy to install, and has low resistance to electrical current. However, copper media is limited by distance and signal interference.

Data is transmitted on copper cables as electrical pulses. A detector in the network interface of a destination device must receive a signal that can be successfully decoded to match the signal sent. However, the farther the signal travels, the more it deteriorates. This is referred to as signal attenuation. For this reason, all copper media must follow strict distance limitations as specified by the guiding standards.

The timing and voltage values of the electrical pulses are also susceptible to interference from two sources:

- **Electromagnetic interference (EMI) or radio frequency interference (RFI)** – EMI and RFI signals can distort and corrupt the data signals being carried by copper media. Potential sources of EMI and RFI include radio waves and electromagnetic devices, such as fluorescent lights or electric motors.
- **Crosstalk** – Crosstalk is a disturbance caused by the electric or magnetic fields of a signal on one wire to the signal in an adjacent wire. In telephone circuits, crosstalk can result in hearing part of another voice conversation from an adjacent circuit. Specifically, when an electrical current flows through a wire, it creates a small, circular magnetic field around the wire, which can be picked up by an adjacent wire.

The figure shows how data transmission can be affected by interference.



1. A pure digital signal is transmitted.
2. On the medium, there is an interference signal.
3. The digital signal is corrupted by the interference signal.
4. The receiving computer reads a changed signal. Notice that a 0 bit is now interpreted as a 1 bit.

To counter the negative effects of EMI and RFI, some types of copper cables are wrapped in metallic shielding and require proper grounding connections.

To counter the negative effects of crosstalk, some types of copper cables have opposing circuit wire pairs twisted together, which effectively cancels the crosstalk.

The susceptibility of copper cables to electronic noise can also be limited using these recommendations:

- Selecting the cable type or category most suited to a given networking environment
- Designing a cable infrastructure to avoid known and potential sources of interference in the building structure
- Using cabling techniques that include the proper handling and termination of the cables

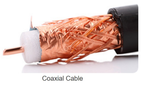
4.3.2

Types of Copper Cabling

There are three main types of copper media used in networking.

Unshielded Twisted-Pair (UTP) Cable

Shielded Twisted-Pair (STP) Cable



Coaxial Cable

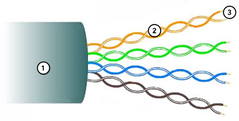
4.3.3

Unshielded twisted-pair (UTP)

Unshielded twisted-pair (UTP) cabling is the most common networking media. UTP cabling, terminated with RJ-45 connectors, is used for interconnecting network hosts with intermediary networking devices, such as switches and routers.

In LANs, UTP cable consists of four pairs of color-coded wires that have been twisted together and then encased in a flexible plastic sheath that protects from minor physical damage. The twisting of wires helps protect against signal interference from other wires.

As seen in the figure, the color codes identify the individual pairs and wires and aid in cable termination.



The numbers in the figure identify some key characteristics of unshielded twisted-pair cable:

1. The outer jacket protects the copper wires from physical damage.
2. Twisted-pairs protect the signal from interference.
3. Color-coded plastic insulation electrically isolates wires from each other and identifies each pair.

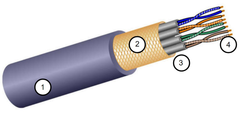
4.3.4

Shielded twisted-pair (STP)

Shielded twisted-pair (STP) provides better noise protection than UTP cabling. However, compared to UTP cable, STP cable is significantly more expensive and difficult to install. Like UTP cable, STP uses an RJ-45 connector.

STP cables combine the techniques of shielding to counter EMI and RFI, and wire twisting to counter crosstalk. To gain the full benefit of the shielding, STP cables are terminated with special shielded STP data connectors. If the cable is improperly grounded, the shield may act as an antenna and pick up unwanted signals.

The STP cable shown uses four pairs of wires, each wrapped in a foil shield, which are then wrapped in an overall metallic braid or foil.



The numbers in the figure identify some key features of shielded twisted-pair cable:

1. Outer jacket
2. Braided or foil shield
3. Foil shields
4. Twisted pairs

4.3.5

Coaxial cable

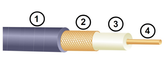
Coaxial cable, or coax for short, gets its name from the fact that there are two conductors that share the same axis. As shown in the figure, coaxial cable consists of the following:

- A copper conductor is used to transmit the electronic signals.
- A layer of flexible plastic insulation surrounds a copper conductor.
- The insulating material is surrounded in a woven copper braid, or metallic foil, that acts as the second wire in the circuit and as a shield for the inner conductor. This second layer, or shield, also reduces the amount of outside electromagnetic interference.
- The entire cable is covered with a cable jacket to prevent minor physical damage.

There are different types of connectors used with coax cable. The Bayonet Neill–Concelman (BNC), N type, and F type connectors are shown in the figure.

Although UTP cable has essentially replaced coaxial cable in modern Ethernet installations, the coaxial cable design is used in the following situations:

- **Wireless installations** – Coaxial cables attach antennas to wireless devices. The coaxial cable carries radio frequency (RF) energy between the antennas and the radio equipment.
- **Cable Internet installations** – Cable service providers provide Internet connectivity to their customers by replacing portions of the coaxial cable and supporting amplification elements with fiber-optic cable. However, the wiring inside the customer's premises is still coax cable.



Coaxial Connectors

BNC | N type | F type

The numbers in the figure identify some key features of coaxial cable:

1. Outer jacket
2. Braided copper shielding
3. Plastic insulation
4. Copper conductor

4.3.6

Check Your Understanding - Copper Cabling

Check your understanding of copper cabling by choosing the correct answer to the following questions.

1. Which of the following attaches antennas to wireless devices? It can also be bundled with fiber-optic cabling for two-way data transmission.
   - UTP
   - STP
   - coaxial
2. Which of the following counters EMI and RFI by using shielding techniques and special connectors?
   - UTP
   - STP
   - coaxial
3. Which of the following is the most common network media?
   - UTP
   - STP
   - coaxial
4. Which of the following terminates with BNC, N type and F type connectors?
   - UTP
   - STP
   - coaxial

Check

Show Me

Reset